AHC Visual Intelligence Hackathon

Real-Time Video Anomaly Detection

AHC runs a one-day visual intelligence hackathon every two weeks. This one is about detecting anomalies in drone video, in real time.

The problem

Drones flying over a city cover highways, streets, parks, railway stations, bus terminals, public gatherings and utility sites, in daylight and at night. Most of what they record is routine, but a small proportion contains events that need a response.

The aim is to detect those events while the drone is still overhead, so that someone can act on them, rather than finding them later in a review of the footage.

Standard object detectors are not sufficient for this. Models such as YOLO identify the object classes they were trained on, but whether something counts as an anomaly depends on context rather than on the object itself. A stationary car is unremarkable in a parking bay and a problem on a highway shoulder. Classical computer vision methods run into the same limitation.

Vision-language models are a better fit, because they are not tied to a fixed set of classes and can be queried in language, which makes it possible to describe an event in terms of its context and to cover situations that were not specified in advance. The difficulty is cost. The large models are too slow and too expensive to run continuously on live video, and inference has to stay cheap enough to run across a large number of drone feeds at once.

The question this hackathon is built around is whether a small vision-language model can do this reliably in real time.

Scope

Some examples of the events of interest:

	Event
1	Traffic congestion
2	Vehicle breakdown, including vehicles stopped on a highway or parked where they should not be
3	A vehicle blocking traffic
4	Accidents
5	Smoke and fire
6	Water logging or Flood
8	Loitering or Suspicious Presence
9	Road spill or debris
10	Fighting or Violence
11	Wrong easy driving

This is not a fixed list. Participants are welcome to cover other events that would matter to someone responding on the ground.

Note that these do not all appear in video the same way. An accident is over in about a second, congestion builds gradually, a stopped vehicle only becomes an anomaly once it has been stationary for some time, and an open drain is a static condition rather than an event. False alarms also matter as much as missed detections, since an alerting system that fires regularly on ordinary activity stops being used.

Constraints

The system must run in **real time** on **limited GPU capability**, so that inference stays economical when it is running across many drones.

Larger hosted models can be used during development, for comparison, or to generate training data, but they cannot be part of what makes the detector work at runtime.

Beyond that, the approach is open. Any of the following are reasonable directions:

- Fine-tuning a small vision-language model on footage of this kind
- Distilling a larger model down
- Pairing a lightweight always-on stage with a heavier verification step
- Training something purpose-built
- Implementing recent published work

What is provided

Unannotated drone footage over urban areas, including night flights, and public benchmark datasets downloaded in advance so that setup does not take up build time. There is enough footage to fine-tune on (Drone, CCTVs and Dashcams), Please Refer to the Dataset Doc shared with you.

Who it is for

- ML and computer vision engineers
- Multimodal AI builders
- Researchers and students working on video understanding, fine-tuning and distilling small vision-language models, efficient inference, anomaly detection or open-set recognition

Participants should arrive with a coding setup and model access already tested, and with a training environment ready if they intend to fine-tune anything on the day.

Schedule

Time	Session
9:00 – 9:30	Breakfast and introductions
9:30 – 11:00	Session on the current state of the art in video anomaly detection, along with related work in robotics and computer vision, and demos from the FlytBase team
11:00 – 06:00	Build
06:00 – 07:00	Selected demos and results

Date: 05 September 2026

Venue: FlytBase Labs or Online